

Lesson 2: Solving Equations with Higher Exponents

Criteria	Exceptional (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding & Strategy	Identifies the appropriate strategy (e.g., roots, factoring, substitution) immediately. Understands potential for multiple roots (real/complex).	Identifies the correct strategy with minor prompt. Understands the main steps to isolate the variable.	Attempts to use exponents rules but struggles to identify the correct overall strategy for solving.	Incorrect strategy applied; demonstrates lack of understanding of exponent laws or inverse operations.
Accuracy & Computation	All steps are mathematically accurate. Correctly applies exponent rules and finds all solutions.	Overall correct, with minor calculation errors. Most roots are found correctly.	Multiple calculation errors. Misses key solutions (e.g., only gives positive root for $x^2 = 16$).	Math is largely incorrect; misunderstands basic exponent rules or algebraic manipulation.
Justification & Steps	All steps are shown logically. Clear, concise justification for steps (e.g., justifying taking an even root).	Steps are shown, but minor gaps in explanation exist.	Steps are hard to follow or missing key logical transitions.	Solution is unorganized and difficult to follow; minimal work shown.
Final Answer Form	Answers are presented accurately in proper notation (e.g., set notation $\{ \}$, exact radical form, or complex numbers).	Answers are correct but not in the most simplified form or correct notation.	Answer is partially correct or missing complex/negative solutions.	Final answer is incorrect, even if some steps were correct.